Aptiv Wind River Telecom Teach-In – Transcript

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Prepared Remarks:

Jane Wu | Vice President, Investor Relations and Corporate Development

Hello, and thank you for joining our event today. Before we begin, I would like to remind you that today's discussion will include forward-looking statements based on our current view of future performance, and that actual results may differ for reasons that we cite in our form 10-K and other SEC filings. A replay and transcript of today's event will be published on our website at ir.aptiv.com. With that, let's get started.

Kevin Clark | President and Chief Executive Officer

Hello everyone, and welcome to the Wind River Telecom Teach-In. Since completing the acquisition of Wind River a little over a year ago, our investment thesis remains on track. As the world becomes increasingly software-defined, we've been leveraging Wind River's technologies and decades of experience to strengthen their competitive position in the telecom, A&D and industrial markets, while also pursuing new opportunities in the automotive market. This has translated into increased commercial traction in automotive, particularly in China with awards from OEMs such as Geely; in A&D with key awards from industry leaders; and in telecom which is our primary focus today.

Building off our Software Teach-In from last fall, we wanted to provide you with a deeper look at what Wind River enables for the telecom industry, which today is addressing a number of the same challenges currently faced by the automotive industry in its transition to the software-defined vehicle. Our partners serving the telecom industry have made significant progress transitioning from hardware-defined to software-defined solutions and Wind River has played a significant role in enabling this transformation. We'll also share how Wind River's technologies and experience from telecom can be applied to the automotive industry both today and in the future.

A software-defined future allows the delivery of more intelligent and responsive solutions enabling the linkage of devices that previously had no way of interacting and creating new solutions in a more connected world some of which you will hear about today from Benjamin Lyon and Anant Thaker from Aptiv, and Avijit Sinha and Paul Miller from Wind River. There's a lot to talk about, so let's get started. I'll now turn it over to Benjamin Lyon, Aptiv's Chief Technology Officer, who will kick off today's discussion.

Benjamin Lyon | Senior Vice President and Chief Technology Officer

Thanks Kevin. Before turning the presentation over to the Wind River leadership team, I'll provide a brief overview of the current state of the telecom industry and Aptiv's software strategy. As Kevin just mentioned, all industries across the world are becoming more connected and software-defined. So what does that mean?

Well, it means just like our smart phones, everything is connected to the internet; software is always up-to-date; products are able to improve and deliver increased value over time; and data derived from this connected world is able to provide ever-increasing insights through the use of AI. This path towards a more software-defined world is not a new trend -- but rather has been building over the last decade and has only been speeding up with ever more powerful and capable networking. We saw this digital transformation start out in Consumer and Enterprise, and are now seeing it expand into mission-critical industries like telecommunications, aerospace and defense, and automotive. And Wind River continues to play a central role enabling the transformation.

Core drivers of value causing this change are up-integration, virtualization, and artificial intelligence. We'll dive deeply into these throughout today's presentation. And as we discussed at CES this year, Aptiv and Wind River continue to be focused on delivering the Right Hardware, the Right Software, and the Right Tools to enable our customers to accelerate and unlock their software-defined futures.

First, let's talk about the Right Hardware. Aptiv and Wind River are delivering solutions that move our customers from relying on many computers that do very little, to just a few computers that can do a lot and are flexible. What makes the right hardware really sing is the Right Software. Wind River's software offerings enable modular and bite-sized over-the-air updates. This is already bearing fruit in a number of deployments of 5G networking globally. And the software-offerings are written so they can be re-used horizontally across multiple projects and multiple applications across industries, and this reduces the cost of scale. Finally, Aptiv and Wind River offer the Right Tools to develop, test, deploy and manage software at scale, all at the highest levels of safety and certification for use in multiple mission-critical industries. But the right hardware, software and tools are not enough. The full benefits of software-defined require the right connectivity, which is being enabled by the deployment of 5G. 5G addresses three connectivity challenges, and those are scale, bandwidth and quality-of-service management.

So let's talk about these. Over the last few decades, the number of connected devices, the volume of data they generate, and the resulting competition for network resources has grown exponentially, and we don't see that trend going away. Customers expect applications to always be connected and continuously updated. This also maps to mission critical industries that have to work every time and all the time.

Pairing the right hardware, software and tools with 5G connectivity enables our customers to unlock new value streams and this is happening now. As mentioned, Wind River is currently supporting the roll-out of 5G across multiple carriers globally. Wind River and Aptiv are also developing the device-side software platform that enables virtualization, bite-sized software-updates, and network quality-of-service management, with all of this being done with a focus on efficiency, minimizing cost, and providing flexibility when choosing the underlying hardware. The combination of these capabilities are very meaningful for the next-generation of connected products.

And we see this being very broad, ranging from telecom to consumer to IoT and to mobility. While 5G will enable many new and exciting business models, in today's discussion, we're going to deep dive how Wind River and Aptiv are providing value across telecommunications and connected vehicle applications. And throughout this discussion, you'll see the unique value of Wind River and Aptiv coming together to bring a comprehensive set of offerings that span from the device, to the edge, to the core, and the cloud.

With that, I'd like to hand it off to Wind River's President, Avijit Sinha, to discuss the telecom industry landscape, where Wind River has a proven track-record of bringing full-lifecycle-solutions to enable some of the most demanding and real-time applications. Avijit, over to you!

Avijit Sinha| President, Wind River

Thanks, Benjamin. As noted, I will spend a few minutes reviewing some of the key industry challenges in the telco market, the evolution of the market, and how Wind River's industry leadership presents further opportunities for 5G deployments.

Beginning with industry challenges. As 5G network coverage expands and more and more devices are connected, telco operators are presented with new challenges. The first is total cost of ownership: As the number of nodes increases, the cost rises exponentially for telco operators. And with extensive scale, the ability to automate, orchestrate and manage feature rollout is critical to achieve a time to market advantage. Last, it has been difficult to monetize new services and achieve a return on investment. Nonetheless the telecom industry has been evolving.

Within this evolving telecom market landscape, the shift from hardware-defined to software-defined networks has been transformative. Traditional network architecture relied on vertically integrated, single-purpose hardware and proprietary software where the service provider would purchase proprietary RAN infrastructure from a small number of vendors. This approach posed challenges across technology, architecture, applications and business considerations. These hardware-defined solutions are proprietary, don't interoperate with other solutions and have high costs for support and maintenance. Furthermore, after an initial competition between one or two suppliers, the service provider gets locked into the selected vendor's solution with no flexibility. Locked into a proprietary solution, service providers find themselves unable to rapidly deploy new network services or applications to generate new revenue streams.

However, now with the rise of software-defined solutions, services providers are able to decouple software from hardware enabling flexibility, agility, scalability and cost-efficiency. These Software Defined Solutions take advantage of two proven technology patterns. Number one, they are based on cloud native architecture that has been proven to scale efficiently at a global scale. Second, these software-defined solutions are built using open source technology that has contributions from thousands of developers and many suppliers and vendors.

This decoupling of hardware and software, adoption of cloud native architecture, and open source technology enables service providers to now choose from a large pool of suppliers, thereby enabling flexibility, choice and freedom from vendor lock-in. Service providers can now respond swiftly to changing customer demands, deploy new services and applications that can enable them to monetize their investment. This critical transformation of the telecom market from hardware to software-defined networks represents a paradigm shift in how the infrastructure is designed, deployed, managed and monetized.

Diving a bit deeper, many operators are now turning to V-RAN, or virtualized-RAN, and Open RAN to tackle 5G's challenges. V-RAN essentially enables virtualization of hardware and decouples hardware from software. So purpose-built appliances are replaced by commercial, off-the-shelf servers which offers a lot more choice. O-RAN essentially enables interoperability between different software and hardware components. So operators can choose different suppliers for different components, thereby eliminating vendor lock-in and enabling flexibility. This enables operators to pool hardware resources, thereby driving down hardware costs. This also enables them to dynamically scale up and down their services to meet peaks and troughs in traffic over a network without needing to overprovision the network. The separation between hardware and software enables both to be updated independently without requiring significant down time. This enables continuous orchestration and management of the network and deployment of new features and applications. All of this provides the operator with significant operating cost savings and the ability to monetize new services.

So why is all of this relevant to Wind River? Wind River is the global leader in delivering software for mission-critical intelligent connected systems. Wind River's platform provides operators with the scalability, high reliability and low latency capabilities essential to support the demanding requirements of 5G applications and services. Wind River's demonstrated expertise in developing and optimizing software solutions for telecommunications infrastructure from the core to edge has enabled many of our customers to adopt and implement V-RAN and O-RAN. Wind River has deployed our platform in over 40K nodes globally across many operators and geographies. ABI Research has ranked Wind River and the number 1 provider of 5G cloud native platforms. In addition to our large scale real world deployments, we have a very strong pipeline of customers and partners working with us to scale out their deployment of 5G using our platform.

Looking ahead, the telecom industry is poised for further innovation. Technologies such as 5G, edge computing, and artificial intelligence will continue to reshape the landscape, driving the adoption of more advanced, software-defined solutions which Wind River is perfectly positioned to enable. With that, I'll turn the presentation call over to Paul Miller, Wind River's Chief Technology Officer, to go over more details on Wind River's technology.

Paul Miller | Chief Technology Officer, Wind River

Thank you Avijit for that overview. Now, let's take a look at the market transformation happening in 5G, and how Wind River is playing a key role. When we talk about 5G Virtual RAN and Open RAN, what do these terms mean? Why is there such a major movement in the service provider industry to adopt these technologies?

First, if you look at the evolution that's happened starting at the left hand side, you can see the legacy approach that's been used for literally decades in the deployment of cellular and telecom networks where products were brought into service providers from major telecom equipment manufacturers that were fully integrated proprietary hardware platforms. Now, of course, inside that platform you have the proprietary hardware itself, but also embedded edge software and above that, the legacy applications. These together were shrink wrapped in a complete solution, as a proprietary hardware appliance, that was delivered from a single company to that telecom operator.

In the next generation product, a shift to an open and multi-vendor approach occurs. This next generation product leverages cloud technology from the enterprise space, as Benjamin explained in his overview. To understand what this means, the industry changed from a proprietary appliance on the left to a cloud-based architecture on the right. This cloud-based architecture is very common in the way that public clouds are built, and how private clouds within enterprises work. In this newer approach, we have commercial-off-the-shelf hardware at the bottom layer from many off the shelf server vendors, then a virtualization technology is applied in the next, or middle layer. In this case that is provided by Wind River Studio, comprised of an open-source software technology founded by Wind River, then commercialized, which contains cloud technologies like OpenStack and Kubernetes. This is a cloud infrastructure that enables you to run a large quantity of server platforms as a fully virtualized cloud that you can manage and operate yourself also known as a “private cloud.” Thus, the service providers are building and operating their own private clouds. Then, on top of all that, the telecom applications depicted here on the top right provide the telecommunications functions, but now as pure software-defined functions without any proprietary hardware.

In summary, the evolution here is from a proprietary hardware based integrated solution on the left to a cloud-based architecture that is heavily software defined. It is this approach that supports virtual RAN and open RAN solutions for 5G deployment, as we’ll see shortly.

Now, why did this happen? Let’s take a look at the business and financial forces in play. Prior to this change, we have some significantly large telecom equipment manufacturers, and a very small number of them, dominating that industry. And they single-source the hardware, the application and infrastructure software, the hardware sparing, the services -- everything that went into the network. Now, obviously the competition at the outset of building a telecommunications network is quite heavy between these few vendors. However, once the service provider selects a particular vendor for their network, they're then locked in for 10 or 15 years or more. This approach restricted the service providers in many ways. It restricted time to market for new services. It limited their ability to reduce TCO. And a single company had a stranglehold on providing technology products into their businesses, thus, limited competition, high prices, high operating costs, and no flexibility. So the movement to the right hand side was actually driven more by financial, commercial and time to market considerations than any desire for a new technology.

So, how did these business considerations produce a new approach? The attributes that you see on the right-hand side drive a highly open, competitive business model. At the lower right side, we have a hardware layer based on commercial off-the-shelf hardware provided by companies such as Dell Technologies, and Hewlett Packard as you see depicted. This is a heavily competitive and commoditized environment. This allows the service provider to swap out that hardware if in one particular year in deployment, Hewlett Packard for example might be less costly than Dell. They can swap that equipment out and move to a new provider. This was impossible before, as one was locked into that one vendor who provided a fully integrated custom appliance.

Same paradigm in the right center layer. In the infrastructure software a competitive market exists, in this case between companies like VMware, Wind River, and Red Hat. These are major cloud infrastructure providers that provide the software technology that creates a virtualized platform upon which to host the application software. These vendors compete at their layer. And then finally, the application software at the top, now no longer tied to a specialized hardware platform, but still from the same providers such as Nokia and Ericsson and Samsung. Due to the cloud architecture, now at the push of a button that software can be replaced by software from a different vendor. Thus, this creates a new environment where there's a tremendous amount of continuous competition, for the life of the network, which drives total cost of ownership down, enables a larger and more diverse set of vendors, creating more innovation, competition and faster time to market. So, some interesting changes happening in the telecom industry as a result of this transformation.

Now, to Wind River. How did we enter and take advantage of this market transformation? As a bit of a reminder, let’s review Wind River’s technology portfolio. On the left hand side, we see Studio Developer. This is useful for many different industries, solving the modern software development paradigm. Software development is complicated and Studio Developer significantly accelerates and reduces the cost of software development. Across the bottom of the diagram is our edge based products. These have been used for decades in multiple industries globally including automotive, industrial, aerospace, defense, medical and telecommunications. From the Mars Lander, to commercial and military aircraft, to automobiles, Wind River technology can be found in many mission critical environments.

We're going to focus on the top right-hand area today. This is the portion of our portfolio that we call Studio Operator. This set of assets is employed in running live operating environments. This is where the transition from developing software to operating software occurs. For example, deploying software, maintaining it, providing security updates, patches, and upgrades known as lifecycle management. Almost every device today requires this, from your cell phone to your TV set. For a telecom service provider, this is the very core of what they do, because they operate a network, providing 24/7 internet connectivity, telephony services, emergency 911 services, and many more offerings. Studio Operator is designed to satisfy those needs, and our technology has become the foundation of the global deployment of 5G, Virtual and Open RAN.

Now, why is Wind River's offering significantly superior to other software vendors? 5G offered a technology inflection point where, due to the business changes we described a moment ago, a new solution was required based on cloud technology, at the same time as greenfield deployments were planned for 5G. This allowed Wind River to become a disruptive entrant into the market at the same time there was a transformation in the use of virtualization technology from virtual machines to cloud native architectures. We were able to innovate in this space by founding an open source project with unique attributes. Leveraging Wind River's expertise in mission critical systems, we created a highly reliable, highly secure, and highly scalable solution.

This innovation allowed us to leapfrog existing vendors, that had technologies more attuned to older centralized datacenters. Rather than a centralized datacenter, a 5G network may require 100,000 sites deployed across an entire nation, but operated as a single cloud infrastructure. These new requirements created the opportunity for innovation. A few examples are depicted here. An optimized footprint is particularly important for a service provider deploying 5G, as across the thousands of edge sites it has a tremendous impact on total cost of ownership. Ultra-low latency is important as we deploy at the edge of the network for the 5G RAN, knowing we are running a virtualized radio platform. This is very much a real time application. This became incredibly important, and this is, of course, an area in which Wind River excels. Then, consider operations. Service providers spend billions of dollars in operating their networks. Many features of Studio Operator reduce OPEX and total cost of ownership for the service provider as they deploy and maintain their network. By simplifying their operations, our solution is preferred over other vendors. Thus, a variety of unique and competitive innovations were created in the technology, and that has caused Wind River to overtake our competitors and become the primary system deployed for 5G Open RAN globally.

What have we done that really impacts the service providers’ OPEX and reduces their cost to deploy and operate this network? It's really a factor of three things. First, on the left hand side is automation of operational tasks. This is understanding that the service provider has to deal with a lot of maintenance actions. Automation reduces time and eliminates error from manual operations. Second, in the center column, the ability to deploy and manage the infrastructure itself. This is particularly challenging over a 5G network, where there may be tens of thousands of sites spread across an entire nation. Finally, on the right hand side there is lifecycle management, the ability to deploy an application and manage that application throughout its lifecycle, including upgrades, maintenance events and security updates. These create tremendous operational burdens for the service provider and really impact the availability and performance of their network.

The solution to these capabilities is Wind River Studio, which can be seen here in this demonstration. This is a single pane of glass that allows the service provider to sit in one location and see their entire network. You can see the entire geodetic architecture of the system, tunnel into any site and manage software within those systems, whether it be the infrastructure or the software application, or automation of their day to day operations. What we are seeing in this demonstration is an example of the ability, from a single user interface or single pane of glass to see every asset in a deployed network across an entire nation. The figures you are seeing are the state diagrams that are used to deploy software into many environments, from public cloud to Wind River's technology at the 5G edge. Finally, what you are seeing is a completely automated engine that eliminates manual action in the network reducing security errors, simplifying management, and creating faster time to revenue while lowering costs of operation. The significant reduction in operational costs is one of the primary reasons service providers choose Wind River Studio, while also enabling our customers to completely automate the deployment and management of this infrastructure and the applications that they are deploying.

Now, as we look to the future it becomes even more exciting with the addition of artificial intelligence. As you can imagine, these huge geo-distributed networks are very difficult for a human being to manage, and thus the introduction of artificial intelligence enables a higher level of automation for the service provider. The demonstration that you're seeing here is an artificial intelligence applied to the top of a live system. This virtualized open RAN 5G deployment that we're managing in this demonstration is using an AI much like what you may have experienced with ChatGPT, a large language model, the difference being that instead of being trained on a static set of data, this AI engine is directly interacting with the live network. Without this capability, a user would require hours and even days to manage the network navigating clicks on a graphic user interface, authoring scripts, or perusing manuals. With this capability, simply ask the AI in natural language, are there any certificates that are expired? Any systems that have an alarm? How do I address that problem? Can you give me suggestions for commands to fix it? The AI then responds immediately with answers resulting from direct interactions with the live network, producing large savings in operational costs and enabling lower skilled workers to manage complex systems. As a result, we can provide the ability to interact with an AI in a very natural way for a human being. This is a future capability that we're rolling out in upcoming product releases.

By addressing the challenges of legacy architecture, Wind River Studio has become a very attractive and proven solution for large scale 5G Open RAN and virtualized RAN deployments. Our solution helps customers reduce up to 75% of the cost to deploy and manage their network. And this, combined with the performance improvements we've discussed is a primary reason we've overtaken our competition in this space. Because they're moving to a software-defined network, we also enable them to accelerate time to market. And as Anant will talk about in a moment, it also enables new revenue streams from new applications that a software defined network allows them to achieve.

After all this, what has happened for Wind River commercially? Several years ago, we competed against the large companies mentioned previously for Verizon's 5G network. We won that business and have been facilitating the roll-out of their 5G network for several years now.

That system is very robust as a result of the years of deployment. We are now at north of 40,000 sites today and growing as we support Verizon's build out of 5G across the United States. This is the largest deployment in the world of 5G Open RAN, and for anyone here today with a cell phone on the Verizon network your phone's 5G service is powered by Wind River Studio Operator.

With our deep investments in Open RAN and the unique innovations of our platform as well as the hardening of the system within the Verizon network and the enhancement of many features and capabilities, Wind River's Studio Operator continues to outpace the competition. As a result, we have won Europe's 1st deployment of Open RAN with Vodafone, and continue to deploy that technology today. Recently, Vodafone came forward with objective test results comparing traditional, proprietary RAN performance and KPIs with the cloud based Open RAN. This was done under the same site locations, weather conditions and same timeframe. Open RAN powered by Wind River was found to be superior in all measurements.

And then finally, another example with Elisa in Finland, who had an amazing experience adopting our full technology stack in Studio Operator. As they began to deploy their edge data centers, they found that the time to deployment, as measured by staff hours to commission the site, including preparation, full installation and testing, was reduced by 90% as compared to doing the very same tasks without Wind River's solution. This is a simply massive OPEX reduction enabled by Studio Operator. They also found that the overall deployment time was reduced by 50%. And then, obviously, because of all of this automation, quality also significantly improved. The automation brought in additional benefits where multiple tasks can now be executed, and they're always executed in exactly the same manner. So, 3 exciting customer examples there, with Verizon, Vodafone, and Elisa benefiting from our Studio Operator offering.

As Benjamin introduced earlier, we are currently transitioning from 4G to 5G today, with a massive deployment of 5G network infrastructure ongoing. Note the predictable evolutionary cadence for the cellular generations. As we look to the future, 6G is coming, and we are headed for deployment beginning in the 2030 timeframe. The interesting transformation that we're seeing here is that all the generations up to 4G were physical appliance-based hardware. 5G can be deployed with a proprietary appliance, or you can use virtualized Open RAN architecture. However, the benefits to total cost of ownership, time to market, flexibility and maintenance of the network have really proven themselves out as shown by the scale and maturity of Verizon's 5G network. This has caused the standardization for 6G to be a 100% virtual architecture.

Thus the investments we are making, and the market traction and leadership position we have in 5G Open RAN, positions us well for the transition to 6G. We expect the TAM to grow significantly as the market becomes 100% focused on the software defined approach for deploying 6G networks. Now, there are many more exciting things coming because of these technology transitions. Here to talk more about what we'll be able to do in the future to enable new revenue generation with telecommunications, and how this also ties to the automotive industry, is Anant Thaker, who is Aptiv's Chief Strategy Officer and Senior Vice President, Product.

Anant Thaker| Chief Strategy Officer and Senior Vice President, Product

Thanks Paul. First, I'd like to highlight the significant parallels between the technologies reshaping the telco market and the similar transformation in its early stages in the auto industry.

On the left, we see what Paul just walked through: what were previously fully integrated, proprietary solutions in teclo are now open and flexible, with a cloud-native platform such as Wind River Studio running on commercial off-the-shelf hardware that can come from multiple vendors. This hardware hosts application software built from ran vendors that can be managed in an optimal way with Wind River Studio Conductor.

In the auto industry, we have repeatedly stressed that to enable software-defined vehicles, we similarly need: optimized hardware – like Aptiv's advanced domain controllers and smart vehicle compute solutions that are flexible and work with multiple SoC providers. On top of that, we need a software platform such as Wind River's VxWorks, Linux, and Helix that can host software applications that are architected with a modern, services-based approach across all levels of functional safety. Then, the software applications themselves such as Aptiv's Gen 6 ADAS need to be actually architected this way, using containers to break apart legacy software monoliths into manageable components that can be more easily developed, tested, and updated throughout a vehicle lifecycle. And finally, the industry needs an edge-to-cloud DevOps platform like Wind River Studio to enable optimized lifecycle management of this software. Importantly, all of these software-defined vehicle technologies can and are being implemented today. In fact, Aptiv's recent Gen 6 ADAS platform award with an EV OEM leverages all of assets shown on the right.

And as we look ahead and 5G network coverage continues to expand, latencies for data transfer between vehicles and the edge cloud continue to decline, and system architectures in both telco and auto become software-defined, we expect several new use cases to emerge that will generate value for multiple stakeholders and open up new business models in this connected ecosystem. And with our full-stack, edge-to-cloud portfolio, Aptiv and Wind River are uniquely positioned to capitalize on these opportunities. Let's walk through some examples. First, the auto industry today leverages telco networks for over-the-air updates and streaming data to and from vehicles. Telco operators often tell us about the challenge they face from the tsunami of devices on their networks that are continuously demanding software updates. And building on what Paul described earlier, Wind River's unique positioning in both telco network infrastructure and vehicles enables the optimization required to address such a challenge in multiple ways. First, using Kubernetes to orchestrate over-the-air updates of containerized software can reduce file sizes and the cost of data transmission. And prioritizing updates that rely on cellular connectivity to off-peak times can help telco operators balance network loads OEMs further reduce costs.

These benefits can be further optimized with cellular-vehicle-to-everything or CV2X. These use cases leverage high-bandwidth network capacity to offload functionality from the vehicle to the edge cloud. This enables OEMs to increase in-vehicle feature functionality while reducing their compute costs. Now let's look more specifically into how this can work in the user experience and ADAS domains.

In User Experience, some graphics rendering and software applications convey critical vehicle information that needs in-vehicle processing, often in real-time. But rendering other graphics such as advanced 3D visualizations can be offloaded to the edge cloud as modern telco networks can deliver the round trip data processing within latency requirements. This is important because graphics rendering often requires GPUs on SoCs to process those workloads, resulting in a higher cost of compute to deliver that functionality. The ability to shift some of these graphics rendering demands from the vehicle to the edge would enable significant savings for OEMs and provide them with greater supply chain resiliency and flexibility.

And in ADAS, while a significant amount of processing has to happen real-time within the vehicle, system functionality can be enhanced by leveraging the edge cloud in ways that have historically been limited by either the right data not being readily available to the vehicle, or too much processing power required to aggregate it. 5G, quality of service management, and edge-to-cloud software platforms can address these constraints. As an example, let's look at the environmental model used by ADAS systems. This is the digital representation of the physical world outside the vehicle including roads, vehicles, and other vulnerable road users. Base environmental models can continue to be generated in the vehicle – but can now be augmented through the edge with enhancements such as real-time traffic and pedestrian information from other sensors. And 5G enables high priority data traffic such as this to have guarantees to ensure it reaches its destination in time. This would enable an enhanced view of the world around the vehicle to proactively identify threats not perceived by traditional sensors and ultimately deliver increased performance at optimal system cost.

Importantly, Aptiv and Wind River are working with telco operators today to bring these user experience and ADAS use cases to market. And they are only a few examples of how the software-defined convergence between the telco and auto markets open up new, value-creating opportunities for which Aptiv and Wind River are uniquely positioned. And with that, I'll hand it back to Benjamin to wrap up.

Benjamin Lyon | Senior Vice President and Chief Technology Officer

Thanks Anant. To summarize, the transition to a software-defined future is happening now across multiple industries and this is being further accelerated by the rollout of 5G. And as Anant just went through in detail, additional opportunities are being opened up like C-V2X applications and network-optimized software updates. To our knowledge, we are the only company delivering software solutions that span from the device to the cloud, across multiple industries. Aligned to this transition, together, Aptiv and Wind River are perfectly positioned to provide our customers with the Right Hardware, the Right Software, and the Right Tools. And this comprehensive edge to cloud offering provides a unique value to our customers in an ever-more connected world. Thanks for your time, and now I'd like to hand it over to Jane to moderate us through Q&A.

Q&A Section:

Jane Wu | Vice President, Investor Relations and Corporate Development

I'm now joined here by our presenters, so let's dive right into Q&A. First question, how has Wind River positioned itself as a leader in the telecom market? Specifically, what is the significance of the 40,000 vRAN and O-RAN nodes already deployed in terms of market share and potential market opportunity? Paul, we'll start with you.

Paul Miller | Chief Technology Officer, Wind River

Thanks, Jane. Wind River has positioned itself as a leader by creating innovative solutions, ideally solving the customer problems in the telecom market. We've been taking share and growing into a market-leading position because of that. The 40,000 nodes that were referenced, Wind River is the market leader. We've won every major deployment of vRAN and O-RAN globally, including Verizon. Our deployment with Verizon is the largest virtualized RAN in the world. Avijit?

Avijit Sinha | President, Wind River

Yes, the global installed base of nodes is actually in the high single-digit millions, but that is traditional RAN or legacy components. As these service providers seek to adopt 5G and roll out modern services, they're going to need the modern technologies that vRAN and O-RAN bring to the table. As that transformation takes place in the telecom industry, the increasing adoption of vRAN and O-RAN will pull in our technology and will become a larger component of the deployed base globally.

Jane Wu | Vice President, Investor Relations and Corporate Development

Thanks, Paul and Avijit. Second question, please explain the components and the $7B+ telecom TAM by 2030. What are the largest drivers, and what do you consider addressable?

Paul Miller | Chief Technology Officer, Wind River

If we start with the U.S. alone, spending from the service providers within the United States is about $30B a year, and about 80% of that today annually is dedicated to 5G infrastructure. That spend number is allocated to all the components required for deploying 5G. That could include radios, core equipment, and the RAN. The $7B TAM that we quoted, represents the TAM for Wind Rivers cloud technology, which is a subset of the total 5G TAM globally.

Anant Thaker| Chief Strategy Officer and Senior Vice President, Product

Maybe just to build on what Paul said, ultimately the key drivers of the TAM are really, the overall size of just the global infrastructure, telecom infrastructure for operators, and just that's represented by nodes, and then the share of that install base that's virtualized, and that's driven by adoption of 5G and then ultimately 6G. And we expect that, of course, to increase quite significantly over time. Then the third is ultimately the economics behind it. Because of Wind Rivers' business model, as you move from the traditional RAN to the virtualized RAN, the economics move from more project-based type of services to SaaS type of business models where they're realizing revenue on a per-node, per-year basis. That's what ultimately drives the TAM.

Jane Wu | Vice President, Investor Relations and Corporate Development

Thanks. All right, next question. What is the competitive landscape in telecom? How is Wind River differentiated versus other software in the market for automotive?

Paul Miller | Chief Technology Officer, Wind River

I guess I'll start with that one. The competitive landscape in telecom depends on what layer in the cloud stack you're at. You could be at hardware, virtualization layer, and applications like we talked about in the presentation. In terms of why a service provider would select our virtualization technologies, such as our competitors like Red Hat, there are several reasons. We've now deployed to reference customers carrying production traffic at high scale. Our competitors don't have that reference. They have absolutely no high-scale production deployments globally.

Wind River also has experienced personnel that have designed and deployed that exact technology that the customers are looking for. Our competitors do not have that experience. Wind River's product solution, if you look at the technology in the product, yields up to a 75% reduction in TCO to deploy and operate this type of system versus our competitors. It's really the combination of these factors, why a service provider would select our solution versus a competitor's. As we shift to the automotive discussion, as Benjamin described, our solutions that have been deployed for decades in industries such as aerospace and defense, are now being used in automotive. Let me hand off to Avijit to talk to that.

Avijit Sinha | President, Wind River

Thanks, Paul. Wind River's software in automotive is differentiated in three ways. Number one, our Studio product. It has a first-mover advantage, because it's the first platform that brings globally distributed teams of developers together to collaborate in real-time and build software at a much faster pace than it has traditionally been done in the automotive industry. It also integrates a wide range of tools that are used by automotive software engineers today. It automates a lot of the manual work that developers have to go through. Then it also enables seamless collaboration, as I said, across globally distributed teams. All of these advantages result in significant reduction in the time to market and the cost of producing software.

Then, when you look at VxWorks, our real-time operating system is the market leading real-time operating system for embedded systems. With the addition of OCI-based, container-based support within RTOS, we are able to bring a lot of capability to the auto industry. The size of updates really goes down, which then reduces the cost of the updates, because the amount of data consumed is less.

Then, when you look at mixed-criticality, our hypervisor, our Linux, and our RTOS combine to allow our automotive customers to really build performance systems using mixed criticality on converged hardware and silicon. So all of these three advantages together result in a lot of advantages for our automotive customers as they seek to adopt modern practices to develop software for the software-defined vehicle.

Anant Thaker| Chief Strategy Officer and Senior Vice President, Product

If I could just maybe add one thing to that, I think actually what you said, Paul, and what you said, Avijit, is really what got us quite interested at Aptiv, because it's bringing the best of the learnings from the telecommunications space around cloud-native deployments for real-time safety-critical applications and bringing that technology into the automotive industry. That's really why we've shifted our own software development and deployment efforts within Aptiv to Wind River Studio and why we're taking our own application software and anchoring that on the Wind River edge platform like VxWorks, Linux, and Helix hypervisor.

Jane Wu | Vice President, Investor Relations and Corporate Development

Next question. Who are the other key industry players in the ecosystem, and how do you expect the ecosystem to evolve?

Paul Miller | Chief Technology Officer, Wind River

I'll start with the telecom side. The key industry players in the ecosystem fit into four general categories. These include the hardware vendors, silicon vendors, virtualization vendors, and application vendors. Wind River has an advanced ecosystem where we partner with silicon providers, such as Intel, to enable best-in-class performance and unique attributes, such as our single-core performance and lower power management. With the application vendors, such as Samsung, we partner to integrate and support sell-with and sell-through of the integrated solutions. With hardware vendors, such as Dell, we provide a fully integrated infrastructure layer optimized for Open RAN. The result of these ecosystem partnerships is reduced complexity and cost to adopt the total solution. Wind River leads in the creation and management of an ecosystem as part of our commercial strategy. There are parallels with the automotive environment, and Anant can talk to that.

Anant Thaker| Chief Strategy Officer and Senior Vice President, Product

Yes, it's very similar, actually. In the automotive space, we have to partner very much with the semiconductor providers and different SOC solutions and Wind River and Aptiv are enabled on several of those. Similarly, we need to work on multiple public cloud platforms. Then in between, the application software can come from Aptiv, it can come from our OEM customers, it can come from other tier 1s, which we may consider competitors. Given how broad that partnership ecosystem is, we need to have open and flexible solutions that can really integrate multiple of those partners to bring the right solutions to our customers.

Jane Wu | Vice President, Investor Relations and Corporate Development

Thank you. As a follow-up to that question, do you see any of the large cloud players trying to move down the stack in competition with Wind River? Avijit.

Avijit Sinha | President, Wind River

Yes, look, the hyperscalers are increasing their investment in data center deployments to grow their compute, networking, and data processing capabilities to support generative AI capabilities. That's really where they make the bulk of the revenue, in cloud and in AI. They're very focused on the cloud and on generative AI. When it comes to the edge, what the edge represents to them is really a socket and a source of data that can go into the cloud and increase storage and compute subsequently. When they look at partnering with the ecosystem, they look at edge players as a source of data and a socket. Therefore, they're very interested in partnering with us.

For instance, now we see Amazon partnering with us to enable studio on AWS, as well as our edge systems connected to the AWS cloud. We'll continue to see this because the hyperscalers will be very interested in partnering with best-of-breed solution providers to pursue specific industries. Our leadership in aerospace and defense, in telco, in automotive, and industrial makes us a great partner for them because our systems, combined with their cloud capabilities, give their customers the best of both worlds, the best-of-breed edge solutions and best-of-breed cloud solutions, with the synergies across connectivity and data, resulting in the outcomes that their customers want across cloud and edge.

Jane Wu | Vice President, Investor Relations and Corporate Development

Next question, how does this competency and experience in telecom translate directly to why Wind River ultimately prevails in auto? Please explain the transferability of your telecom solutions to other industry verticals and the comparable use cases to automotive in particular.

Anant Thaker| Chief Strategy Officer and Senior Vice President, Product

Yes, so maybe I can start because I think I spoke about it in my prepared remarks. I think Paul described quite well what's happening in the telecommunication space, where you're moving from fully integrated proprietary solutions, black box solutions in some ways, right to open flexibility and modularity between layers of the stack. Commercial off-the-shelf hardware providers building RAN applications on top of the Wind River software platform, and then all of that, orchestrated and managed with Wind River Studio. Really all of those same things we expect to happen in the automotive space as you move to software-defined vehicles. And so all of those same technologies are transferable.

We're starting with best-in-class hardware that we need to enable those software applications and that's our advanced domain controllers and SVA, Smart Vehicle Compute. On top of that, we'd be running the Wind River Edge software platform, and that is architected to support OCI containers that's architected to support modern software development languages. All of those things are quite transferable into the automotive space to then enable services-based architectures and modern architectures of application software that sits on top of this platform. Then the development, the deployment, and the lifecycle management of those features. Just as software is managed in deployments and telecommunications, we expect to carry over into the software-defined vehicle. It's really the learnings of what we saw work to solve the use cases and needs in telecommunications, all of that we want to bring and apply into the automotive space.

Paul Miller | Chief Technology Officer, Wind River

We've also seen in the aerospace and defense sector, up-integration much like is happening with SVA in the automotive sector, happening over 10 years ago with companies like Boeing and Airbus, where we've provided the technologies that allow them to up-integrate and virtualize their compute modules exactly the same way that the automotive industry is making that transformation today with software-defined vehicles.

Jane Wu | Vice President, Investor Relations and Corporate Development

Thank you. Next question. Are there tech and/or operating synergies between telecom and other end markets such as automotive and aerospace?

Paul Miller | Chief Technology Officer, Wind River

Yes, so as Benjamin described in his introduction, we're seeing technologies such as cloud-native virtualization, over-the-air updates, up-integration, and software-defined systems being reused from other markets. This really started in the enterprise market. They adopted those technologies first, and then they cascaded down into the telecommunications and aerospace sectors. Now they're, as I described a moment ago, falling into the automotive sector as that system is modernized. This really creates tremendous synergies that we're seeing in the technologies Wind River provides, where that solution is used by multiple industry use cases. The benefit of each industry is that they jointly consume the technology in aggregate and benefit from the greater use of that technology in the commonality.

Finally, as the common software-defined future emerges, it enables a future with advanced applications such as C-V2X, enabled by the use of these common technologies like Anant presented in his prepared comments. Thus, the technology and operating synergies are providing benefits to customers due to reuse across many verticals.

Jane Wu | Vice President, Investor Relations and Corporate Development

Are you seeing the pace of adoption for automotive similar to that of the telecom industry?

Benjamin Lyon | Senior Vice President and Chief Technology Officer

I'll take that and maybe Anant, you can jump in. First, adoption is almost never linear, and it really starts with first movers. We see that with Verizon in telecom, and like Kevin was talking about on the software-defined vehicle side, in China with Geely. Then also, the telecom deployment of 5G really opens up a whole new suite of applications for connected vehicles, which we think will drive further additional value. Anant, maybe some additional color?

Anant Thaker| Chief Strategy Officer and Senior Vice President, Product

Just to build on what you said, Benjamin, I think we are seeing adoption accelerate similar to what we see in the telecommunications space. As you mentioned, it is driven by those early adopters and first movers that are first to move to software-defined vehicles. In fact, all of our commercial pursuits in ADAS and in-cabin user experience, we're anchoring those on the Wind River software platform at the edge, as well as leveraging Wind River Studio. Those are really catalysts that are driving adoption in the market. As network coverage for 5G expands and synergies evolve between automotive and telecommunications world, that's only going to further accelerate the adoption of all of these new use cases and opportunities.

Jane Wu | Vice President, Investor Relations and Corporate Development

What is Wind River's revenue today? What is the percentage from telecom and auto? What is their relative growth through 2030? Similarly, what is the split by region, and how will that evolve?

Avijit Sinha | President, Wind River

I'll take that. Wind River's revenue is about $500 million annually today, and about 20% of that is from telecom. Auto is a smaller percentage but expected to grow at a larger rate over time as we partner with Aptiv and bring our solutions to market. We expect Wind River's revenue to really grow at double-digits over the next few years. In regards to regions, North America today is our largest region, but we expect a lot of growth in China, in particular in the automotive space with Aptiv. With Telco, we are expanding into the APAC and Latin markets as well. There's going to be growth globally across regions and across industries as well.

Jane Wu | Vice President, Investor Relations and Corporate Development

Thanks Avijit. Can you help us better understand Wind River's sales cycle and revenue build within telecom? What is the mix of Wind River's revenue generation, i.e. recurring licenses versus services versus one-time, edge software versus Wind River Studio, and how does this mix differ between telecom versus auto?

Paul Miller | Chief Technology Officer, Wind River

I guess I can start, and maybe Avijit you can follow up. The sales cycle at Wind River for telecommunications is typically a 12 to 18-month cycle. It's pretty long, lab trials and qualification, network engineering, etc. On occasion, we see both shorter and longer sales cycles driven by any number of factors from budget availability or a customer's need for production license as they ship products containing our software. Maybe you can comment on the revenue splits.

Avijit Sinha | President, Wind River

Yes, approximately 20% of Wind River's revenue comes from our services offerings and while Wind River's edge today is the bulk of our software because licensing and services revenue come through that product portfolio and edge is a highly renewing component of our revenue. We have upwards of 90% renewal in our edge portfolio which is great for our business. The studio portfolio comprised of studio operator and developer today is growing tremendously in telco, in automotive, and other industries, and so we expect to see quite a bit of growth there as adoption of cloud-based solution grows. So I think, in aggregate, we'll continue to see double-digit growth as I mentioned earlier, in our portfolio and the aggregate revenue for the company.

Jane Wu | Vice President, Investor Relations and Corporate Development

Can you give us a specific example of Wind River's value proposition in telecom, i.e. cost savings, improved reliability and performance, new business models?

Paul Miller | Chief Technology Officer, Wind River

Yes, sure. Glad to do that. We've been living this the past few years. Our value proposition in telecom is really based on several factors. These factors directly contributed to the examples we gave. For example, the commercial winds at Verizon, Vodafone, Elisa, and Telus. We start with a differentiated product that really enables the lowest possible acquisition cost for the customer. That's typically 75% lower cost to deploy. Then, as the customer starts to move to operate this environment, automations and operation features that directly impact the OPEX and personnel required for deployment and ongoing support. As we mentioned in the Elisa example, savings of 90% and 50%. Also, industry-unique experienced personnel to support this and service the customer, we have the only personnel in the industry with those capabilities. Wind River is also the only company with credible production references that scale globally. Then, we offer a full technology stack, including cloud infrastructure, analytics, and automation components -- the complete set of capabilities. That is why companies like Verizon, Vodafone, and the others are choosing our technology solutions.

Jane Wu | Vice President, Investor Relations and Corporate Development

Any updates on the new and latest Wind River awards, including in auto?

Paul Miller | Chief Technology Officer, Wind River

Yeah, a couple. In telecommunications, we've had a lot of commercial traction. We have ongoing lab trials with customers in the U.S., Canada, Europe, and Japan. The Telus award at the beginning of the year represents a significant win with a new operator and geographic expansion into Canada. It will be Canada's first Open RAN network that we're deploying there. Avijit, maybe you can comment on automotive.

Avijit Sinha | President, Wind River

Yes, look, as Kevin mentioned in his remarks, we've seen several new awards in China. In our aerospace and defense sector, we've seen several wins this year for certified RTOS systems, for major OEMs and Tier 1s across the defense and commercial aviation space. Anant, maybe you can add some on the automotive side as well.

Anant Thaker| Chief Strategy Officer and Senior Vice President, Product

Yes, I think we have awards in China, for example, with Geely. I think most recently, the Gen 6 ADAS award that we have is going to be anchored on the Wind River software platform and leveraging Wind River Studio. We're seeing a lot of commercial traction.

Jane Wu | Vice President, Investor Relations and Corporate Development

Thank you. This is the final question. What are Wind River's goals over the next 12 to 24 months? What innovations or developments can be expected from Wind River in the next few years?

Avijit Sinha | President, Wind River

Yes, our growth strategy is essentially based on three focus areas. Number one, we'll continue to develop new products. For instance, we are going to be introducing a new Linux offering for the enterprise market. The second thing is we will take our existing products into new markets, both geographically as well as across industries. As an example, our Wind River cloud platform will now be brought to other industries as well, where there is tremendous receptibility for that. Then lastly, we will be expanding the scope of our telco products geographically across Latin America and APAC. Furthermore, with AI, we are going to be enabling AI across our product portfolio. In Studio Developer, along with AWS, in fact, we've enabled that in Studio Developer for increasing developer productivity. In Studio Operator, we are employing large language models and generative AI to increase automation in the network. Then we'll be enabling AI at the edge through our runtimes and frameworks on our VxWorks and Linux platforms for customers to enable AI at the edge.

Jane Wu | Vice President, Investor Relations and Corporate Development

That concludes the Q&A. Any final comments, Kevin, team?

Kevin Clark | Chairman and Chief Executive Officer

Listen, hopefully with these presentations, you're as excited about Wind River as we are. We think, as the team said, there's tremendous opportunity within the telco space, within the A&D space, within the industrial space, all of which translates into what's going on in the automotive space. Each and every day, we interact with our customers, we interact with our supplier partners, and when you look at the overall Wind River portfolio, bringing that to bear into the automotive industry. We're confident we'll bring revenues for Aptiv, but also we'll solve significant problems and challenges for our customers. We're very, very excited about Wind River, very pleased with the progress to date, and very excited about what we're going to see in the future. Thanks for joining us today!

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